

Making Your Repository More Accessible to SI

NIAID Blueprint Webinar Series

November 6, 2026

SAN DIEGO SUPERCOMPUTER CENTER

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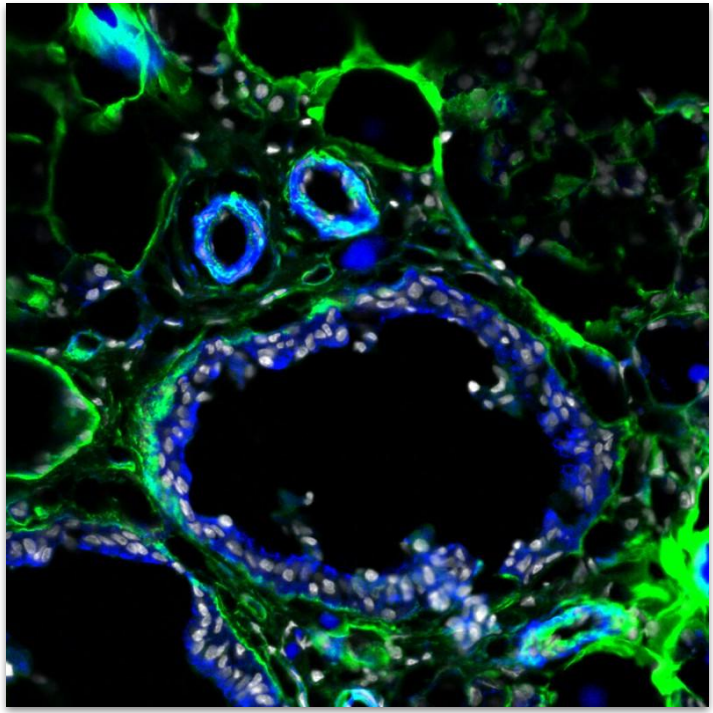
Knowledge Motifs LLC
Mapping sensible data relationships

Agenda

The main themes we will be going through:

1. Scope
2. Why now?
3. Who visits?
4. Be what?
    a. Be reachable
    b. Be readable
    c. Be callable
    d. Be meaningful
5. Evaluation
6. Closing Notes

[Supplementary materials for: Making Your Repository More Accessible to SI]

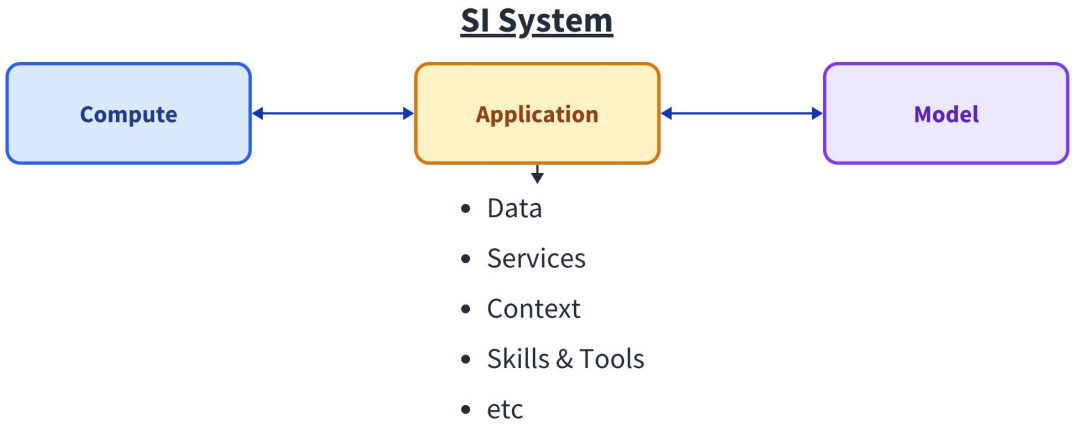


Peekaboo human parotid glands
Images courtesy of the NIH Image Gallery



← Peekaboo human parotid glands after eating leftover Halloween candy, courtesy Google Gemini

Scope



This is a big topic! We have a lot to cover…

This talk is not about:

- Building/coding agents
- Building or leveraging a harness
- Selecting or implementing various models
- Compute environments for agents
- ***Important***, we will also not focus on state of the *data* itself in support of SI or ML use cases.

One common way of thinking about an SI enabled environment is in the three components in the figure above.

This talk is about:

Facilitating SI workflows and agents through:

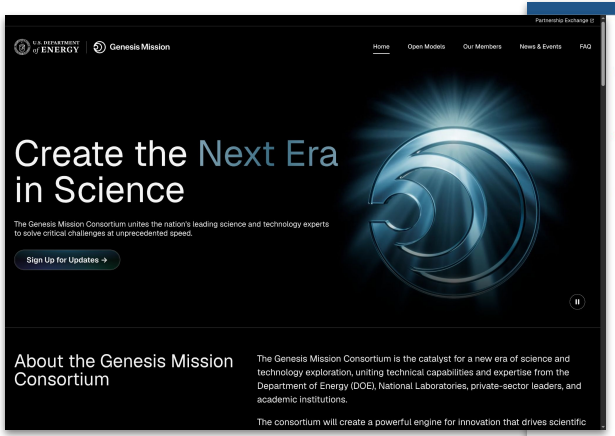
- Repository configurations
- Files & resources (metadata)
- API and MCP services
- Semantic layer components

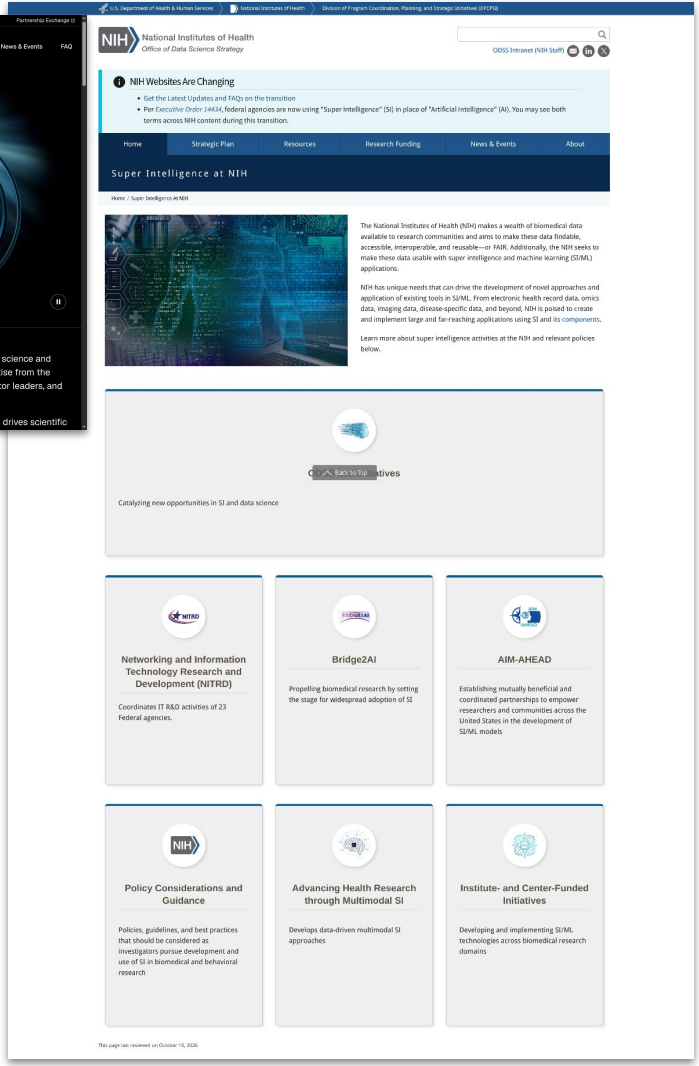
*For this talk, we are focused on those **assets we provide to an application/workflow** from a repository, these include:*

- Data
- Services
- Context
- Skills and Tools
- etc

Why do this?

Growing momentum behind leveraging SI to accelerate science workflows.





NIH Bio Genesis Mission: $1.2B+ in aligned FY26–27 funding to halve discovery-to-patient timelines within ten years.

Bridge2AI Expansion: $130 million over four years for Stage 1, and Stage 2 was approved on January 29, 2026.

Predictive Biology Partnership: On October 7, the DOE/Biohub announced $1.8 billion in funding, data, computation, and measurement technology, with Google DeepMind, Meta, and Isomorphic Labs joining. Biohub will work with NIH to standardize these datasets for AI model training.

National Genesis Scale: $5B+ federal cross-agency funding 278 projects across 342 institutions in the first Genesis Mission round

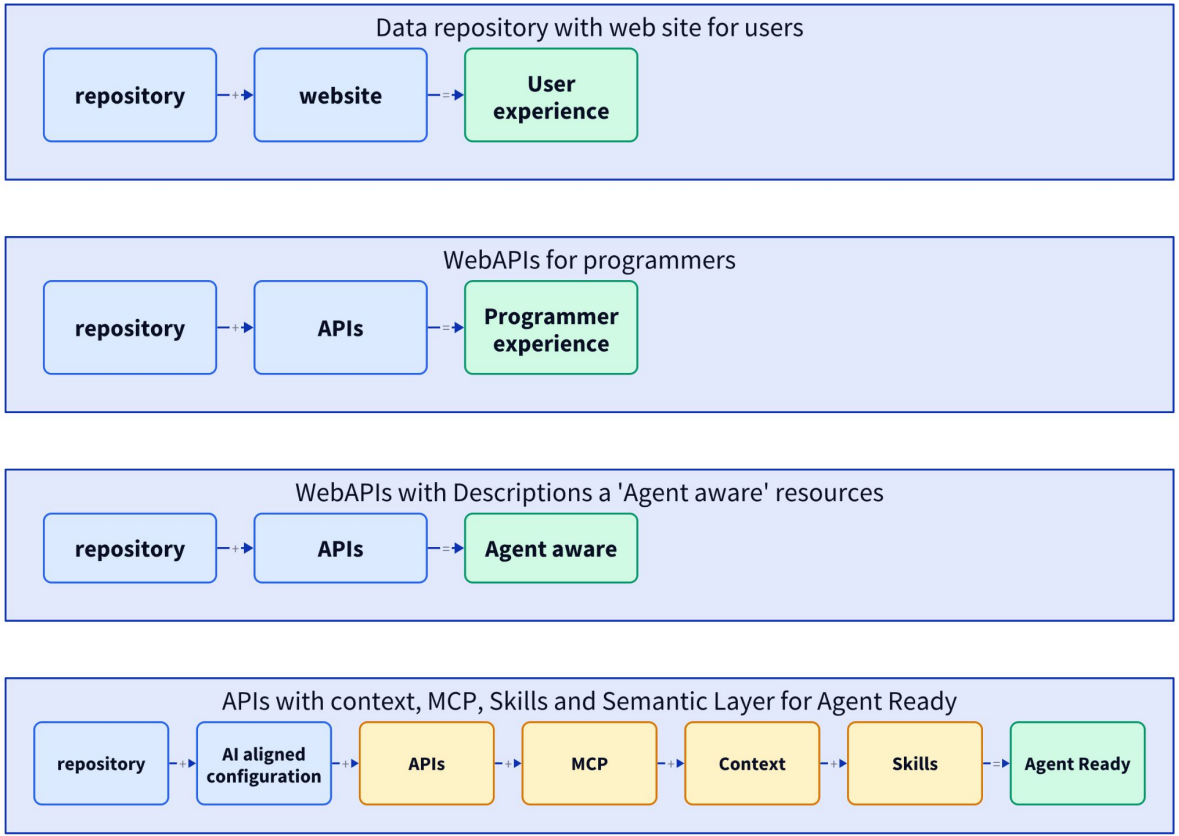
Why do this? *Infrastructure is becoming application*

*From **data repository** to **knowledge system***

The swim lanes to the right show example flows for both humans and agents.

Different approaches for different consumers.

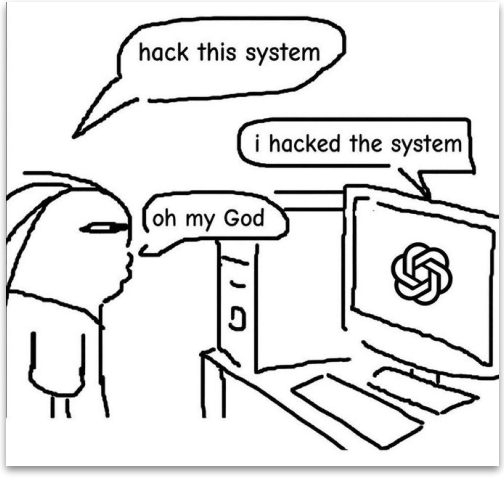
SI Workflows are both learning to mimic humans in workflows and benefiting from focused approaches



Why do this? *Let's raise some concerns up front*

Agents do not have agency…

They are ***acting on the behalf***, and under the direction, ***of your user community*** (or someone).



Blocking may not stop
SI workflows will still search for answers, potentially finding stale context or leveraging blind scraping that may strip out your license, DOIs, and variable definitions.

Rights & access
Who can view or use your data and under what specific conditions.

Attribution & authority
How to ensures that creators, contributors, and organizations receive proper credit for their work. What are standardized ways to verify the original source and clearly outline how the content should be cited.

Lineage & IP protection
Tracking the history of your data to show exactly where it came from and how it has changed and can help address integrity.

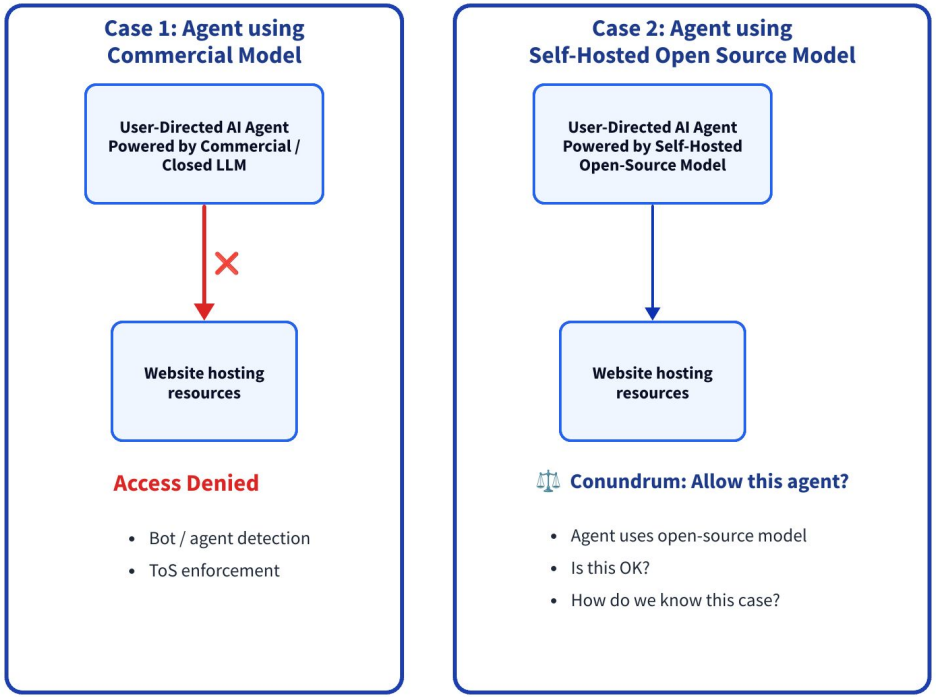
We will revisit these later in the presentation

Visitor Classes: *From the Why to the For Whom*

Trainer crawlers:
Looking for training data (OpenAI, Anthropic, etc)

Index builders:
looking for citable resources, the catalogers that provide value add in terms of discovery and novel search.

Organic requests:
Based on user driven sessions/queries, either in person or via ***"agent visits" on their behalf.***
This agent my even have my credentials.

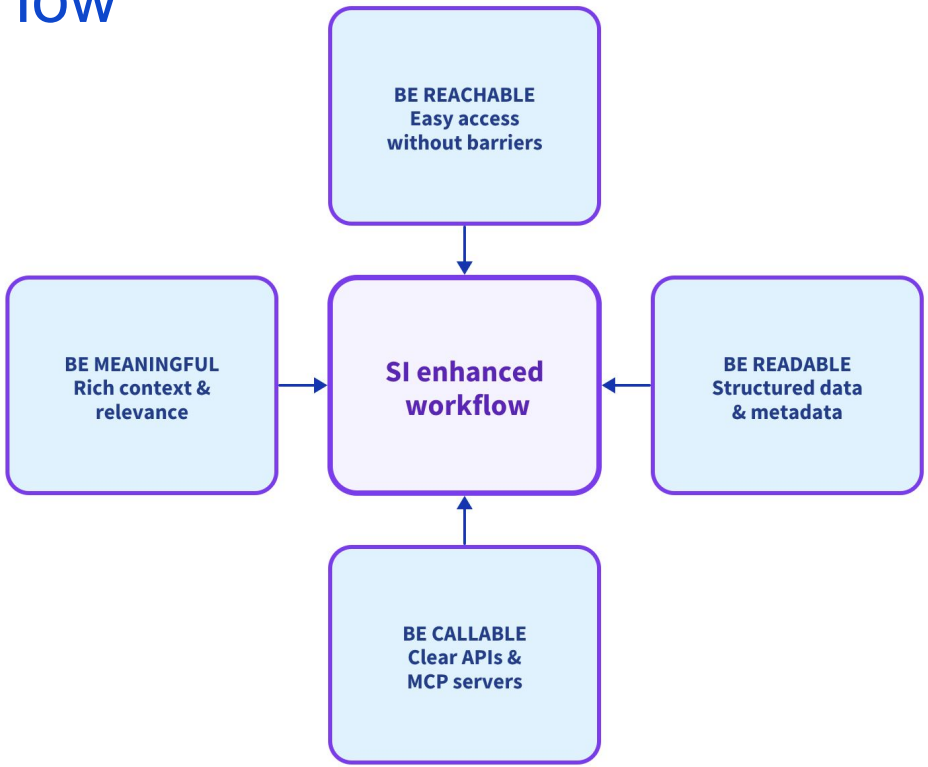


Knowing the model behind your user is very difficult

From Why & For Whom, to How

The rest of this material is broken down into the following four foundational facets.

- ***Reachable*** *(network accessible)*
- ***Readable*** *(expressive resources)*
- ***Callable*** *(dynamic/service reads)*
- ***Meaningful*** *(context)*



Be Reachable

Configuration:

Some sites, or providers, detect and filter automated access, so check for that if using a service provider

Server and Proxy Configuration Notes:

- **Reverse proxies** for example via tools like Kong or Caddy (with module) and many others.
- Load balancing and caching (always a good idea)
- *Http code 429 (**Too Many Requests**)*
- ***Retry-After**, typically goes with 429*

The agents might try to ignore this, but you're taking the high road, and you can then ban them.

+1,700%

Growth in daily SI agent requests over one year.

>50%

Non-human traffic on major CDNs.

https://www.forbes.com/sites/jonmarkman/2026/10/01/cloudflare-says-ai-agent-traffic-grew-more-than-1700-in-the-past-year/

Be Reachable

robots.txt

What is it?
A file named "robots.txt" at the root of your web server.

What does it do?
Messages agents (software accessing your site) with your declarations about policy for access. *(whether they follow the guidance, is up to them)*

The key components to a robots.txt are:

- User-agent
- Allow directive
- Disallow directive

Simple example:

```
User-Agent: *
Disallow: *.gif$
Disallow: /example/
Allow: /
```

There some additions like *Crawl-delay* and *Sitemap* which are not part of the spec, but honored by some agents.

Emerging interest in parameters like:
`Content-Usage: train-ai=n`

Might also be placed in HTTP headers

Be Readable

sitemap.xml

What is it?
A file named "sitemap.xml" at the root of your web server or someplace defined by robots.txt.

What does it do?
Provides a means to express the tree of links in your website.

Predates SI as a means to express site links and structure

Sitemap.xml

You can leverage the XML schema defined at: https://sitemaps.org/protocol.html

You can break out targeted sitemap.xml via the robots.txt file and express them for specific user agents.

Note: Still in use?
Sitemaps are still in use, though some groups will simply crawl a site anyway as the reliability of up to date sitemaps varies by provider.

Be Readable

llms.txt (V2 came out Aug 2026)

What is it?
A file named "llms.txt" at the root of your web server.

What does it do?
Provides links to resources in a format more aligned to SI workflows

llms.txt specification at: https://llmstxt.org/

- adds a standard way to point from each page to a Markdown version (`rel="alternate" type="text/markdown"`)
- (`rel="describedby"`), as HTML link tags or HTTP `Link:` headers. It also lets a llms.txt sit in a subfolder and cover the pages under it.

Tends to see llms.txt more in documentation sites and when feeding code agents

Content negotiation and clean HTML 5 content (with semantic headers) and/or Markdown may address similar goals llms.txt addresses

Be Readable

Signposting and FAIR Signposting
https://signposting.org/FAIR/

Mature, low-cost
Let agents and harvesters reliably navigate from landing pages to PIDs, metadata, content, and licenses without brittle scraping or heavy JavaScript.

Lives either in the HTML or in the HTTP headers. The later is very lightweight and fast, though technically harder to implement.

Widely adopted at the platform level

- Dataverse since ~5.14,
- DSpace,
- InvenioRDM/Zenodo, and others).

Typed HTTP Link headers (and/or <link> elements / Linksets) that declare;

- cite-as
- describedby,
- item,
- license,
- author,
- etc.,

Be Readable

Javascript Rendered Pages

- Many agent fetches may not execute Javascript, so information placed in the DOM at load time in a browser may be missed by AI. However, SI can leverage tools (in memory browsers for example) to deal with this.
- This could become a server load issue with high agent use too

Markdown Conversion

- You may wish to start thinking about how a page looks in markdown. Some content distribution paths now may serve HTML as Markdown if the agents request prefers *Accept: text/markdown*
- Markdown is the current preferred format. Be careful about large PDFs or things like docx that might require an agent to transcode the content.

Be Readable

SKILL.md (https://agentskills.io/)

- Expose your workflow and/or API usage patterns and CLI tools as an agent skill
- Can be a **simple markdown** document **OR** can include
  - *Tool calls (local CLI tools may require prior installation)*
  - *Service calls (API or MCP)*
  - *Invoke workflows*

Package structure:

```
skill-name/
   SKILL.md
   scripts/            # executable code
   references/         # docs loaded only when needed
   assets/             # templates, icons, etc
```

YAML frontmatter

```
---
name: my-skill
description: What the skill does AND when to use it. Use
this whenever the user mentions X, Y, or Z, even if they
don't explicitly ask for "X."
compatibility: (optional) required tools or dependencies
---
```

Markdown body

- Write in the imperative. “Read the input file, then…” works better than describing what the skill “can” do.
- Explain the reasons behind rules.
- Define output formats explicitly (include a template)
- Include examples.
- Point clearly to any references

Be Readable

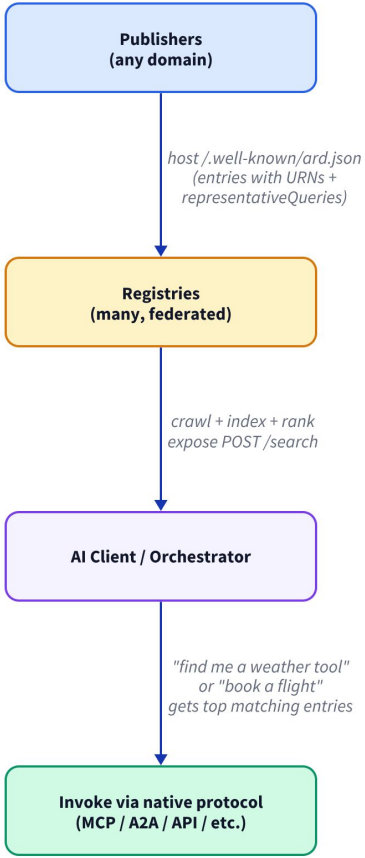
Caution: bleeding edge



Agentic Resource Discovery Specification
(**ard.json**) https://agenticresourcediscovery.org/spec/

What is it?
A JSON file named "ard.json" hosted at:
https://yourdomain/.well-known/ard.json

What does it do?
The Agentic Resource Discovery Specification (ARD) defines how agentic resources are described, discovered, and searched across federated networks.



Be Readable

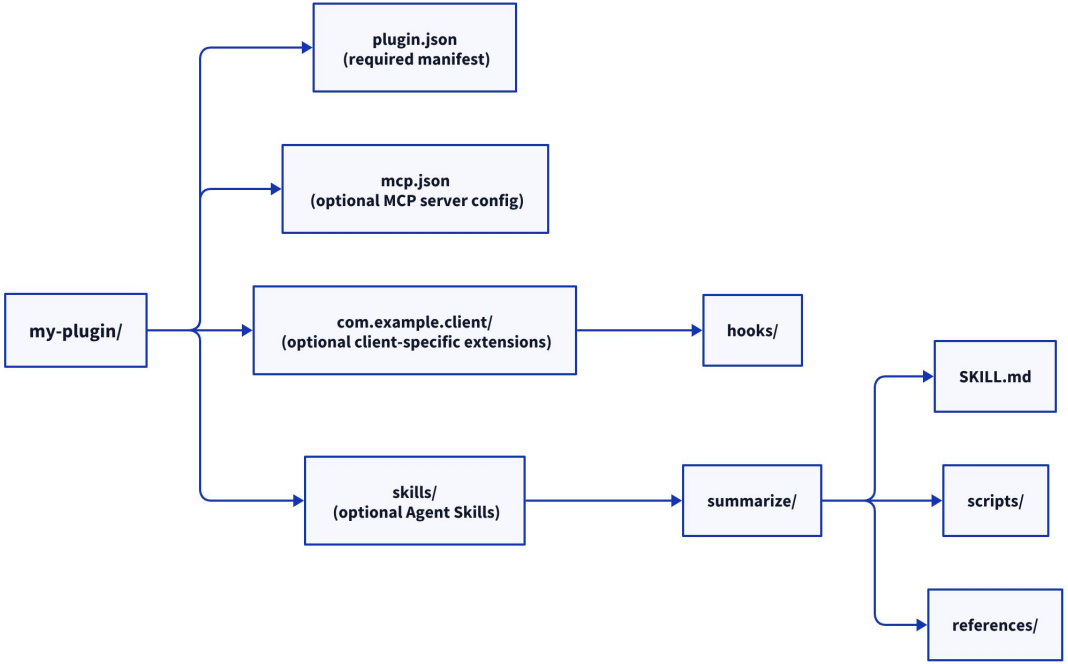
Caution: bleeding edge



Agent Plugin (https://agent-plugins.org/)

What is it?
A packaging format for skills and MCP services.

What does it do?
From the documents: "Agent Plugins is an open, vendor-neutral standard for packaging reusable components into portable plugins…"



Be Callable

A previous webinar addressed APIs:
[API NIAID Webinar Slides 033126]

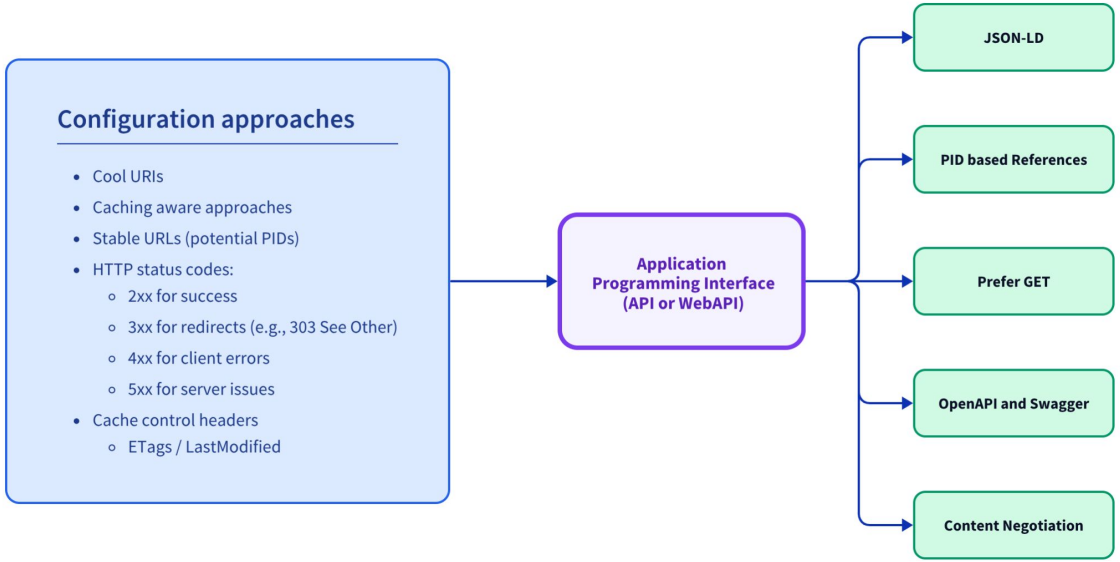
See also the Blueprint:
https://doi.org/10.5281/zenodo.17161561

Leveraging Web Standards for Robust, Scalable API Services

APIs best practices are well established. In terms of "SI aware" APIs some key value points are:

- Good documentation, above all
- Leverage web architecture, status codes, stable URLs, etc
- Connect response object to a well described schema or data model

APIs / WebAPIs



NOTE: **This material is focused on Metadata**, not data resources, so items like JSON-LD and prefer GET are not meant for **your data resource which are likely driven by domain use cases and potential security and access control issues.**

Be Callable

MCP (Model Context Protocol):

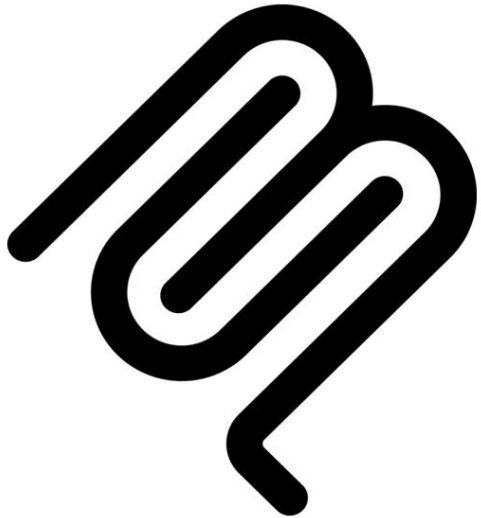
- ***https://modelcontextprotocol.io/docs/2026-07-28/getting-started/intro***
- Governed by the Linux Foundation's Agentic AI Foundation
- Default registry at: https://registry.modelcontextprotocol.io/

Unpopular opinions (so please only blame me):

- MCP is not required for you to address "SI-Ready"
  - Material from Reachable and Readable sections along with a well documented API (ie, OpenAPI/Swagger) goes a LONG way
- MCP alone is not an "SI-Readiness plan"
  - You still need policy, network architecture, context, etc.

That all said, yes, MCP is an important topic to know about!

So let's dive in!



What was the purpose of MCP?:
Agents can get confused navigating APIs, *MCP provides a **shared convention** for a wide range of clients to leverage.*

Be Callable

Recall that MCP has more than just tools, and more than your APIs in a new wrapper

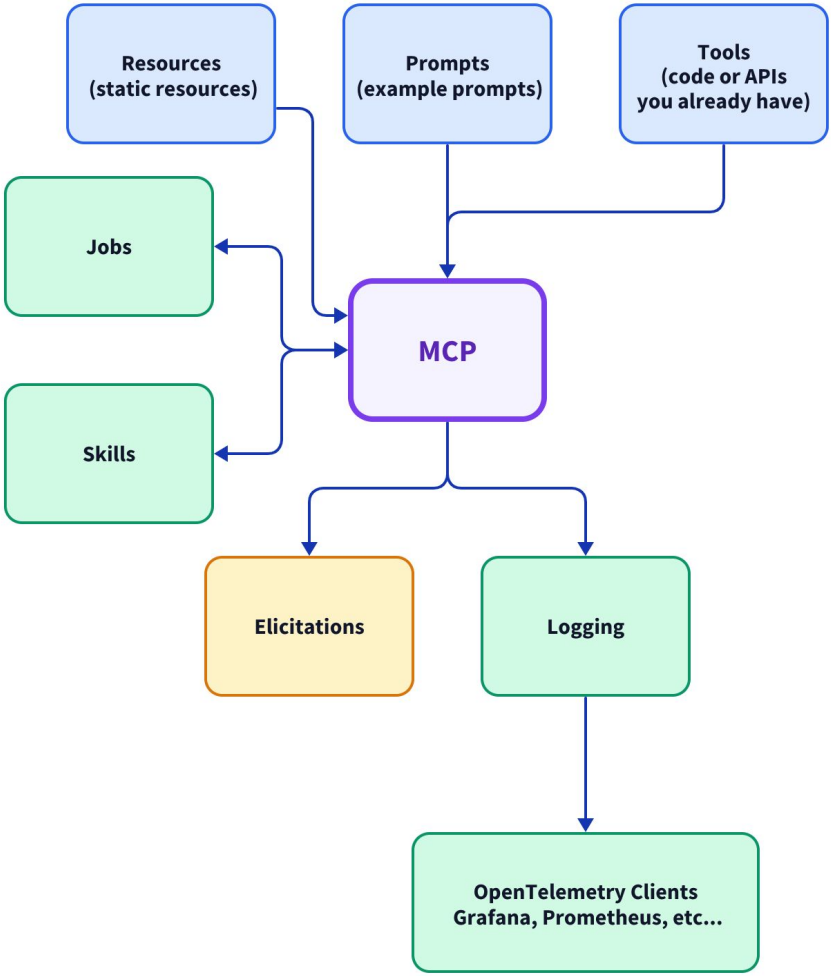
Primitive	Controlled by	Purpose	Discovery / Use	Examples
Tools	Model	Executable functions the LLM can call (may change data, call APIs, run code)	`tools/list` → `tools/call`	Search flights, create calendar event, run a query, send a message
Resources	Application (host)	Read-only data/context identified by URI (files, schemas, docs, DB records, etc.)	`resources/list`, `resources/templates/list` → `resources/read`	File contents, database schema, policy document, calendar data
Prompts	User	Reusable message templates / workflows (often exposed as slash commands or menu items)	`prompts/list` → `prompts/get`	“Plan a vacation”, “Draft a reply”, “Summarize my meetings”

Be Callable

… and beyond Resources, Prompts and Tools

MCP is a hub between data, services, context and more to clients.

*NOTE: Reflect on where other assets discussed in this slide deck **map into MCP as a delivery mechanism,** and* you may find you can share the same resources through **multiple approaches**.



Be Callable

MCP: Several technical updates, some really are important (details)

March 25	**Upgrades system security to modern authentication standards** and improves how continuous data flows over the network. It also adds new ways to clearly define how automated tools should behave.
June 25	Standardizes how servers **handle user permissions** and requires tools to output data in a **more predictable, structured format.** It also cleans up the system by removing outdated methods for grouping requests.
Nov 25	Introduces a **smoother process for applications to verify their identity** and ask users for permissions one step at a time. It also adds user-friendly customization options, like tool icons and standard naming guidelines.
July 26	Greatly simplifies server management by removing the need for the server to remember past interactions **(making it "stateless").** It also streamlines the core software by moving complex background tasks into optional add-on extensions.
Sept-Oct 26	Finalizes the new **"Skills" extension a**nd officially rolls it out in standard developer kits. This allows creators to easily **build and share more advanced plugins** for the system.

Overflow: see more in supplements



Be Callable

MCP: Authentication and Authorization

This is a large topic, both here and elsewhere in the talk. I have provided more information for those interested in the supplemental materials

Agents need a machine-readable way to authenticate and be authorized before they can call tools or retrieve resources. **MCP authorization (2026-07-28 spec)** treats the MCP server as an OAuth 2.1 **resource server**:

- On unauthenticated request → 401 + WWW-Authenticate pointing to Protected Resource Metadata (RFC 9728).
- Client discovers the authorization server, registers (Client ID Metadata Document preferred), runs authorization-code + PKCE, and requests a token bound to this specific server via **resource indicator** (RFC 8707).
- Token is presented as Authorization: Bearer on every request; server validates audience, issuer, and scopes.

Scopes enable least-privilege and step-up (e.g., mcp:resources:read vs. broader access). Server can challenge with the exact scopes needed.

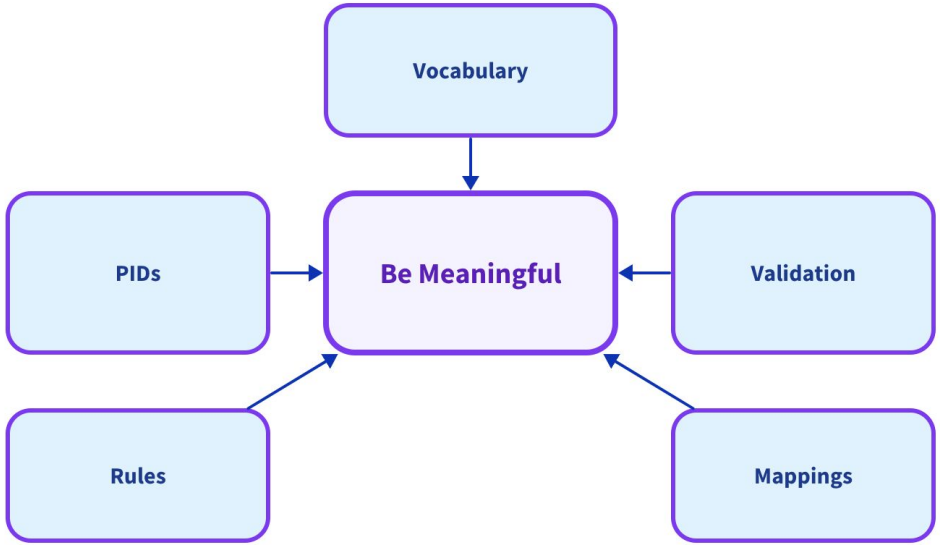
Result: ***agents can act on behalf of an authorized user*** (or use client-credentials where appropriate) without long-lived API keys or manual login. (WebMCP may make this even more complex)

Be Meaningful

Structured knowledge is the most reliable way to reduce runtime schema hallucinations.

Express the context of the data in ways that also **leverage persistent identifiers** and **structured metadata**.

*This aligns with the [NIAID Blueprint] recommendations, and provides important usable context to the mod*els.



Be Meaningful

PIDs
A persistent identifier gives each thing one stable name an agent can cite and follow.

Popular PIDs include

- DOI or accession for data
- ORCID for people,
- ROR for organizations
- RRID for resources
- RAiD for projects

See also:

- Blueprint Series: Introduction to Identifiers (YouTube)
- ***Dec 2026 Webinar***

- **Disambiguation** — uniquely identifies entities (e.g. ORCID for people, ROR for organizations, DOI for outputs).
- **Resolution & citation** — reliably resolves to a landing page or metadata record and supports persistent citation/attribution.
- **Metadata association** — carries or points to rich, structured metadata.
- **Graph linking** — enables the construction of a **PID Graph** in which PIDs are nodes and typed relationships (creator-of, funded-by, cites, version-of, etc.) are edges. This is the model promoted by the FREYA project and operationalized in services such as DataCite Commons and the OpenAIRE Graph.
- **FAIR support** — underpins Findable (F1, F3), Accessible (A1, A2), Interoperable, and Reusable principles by providing stable, resolvable anchors and machine-actionable links.

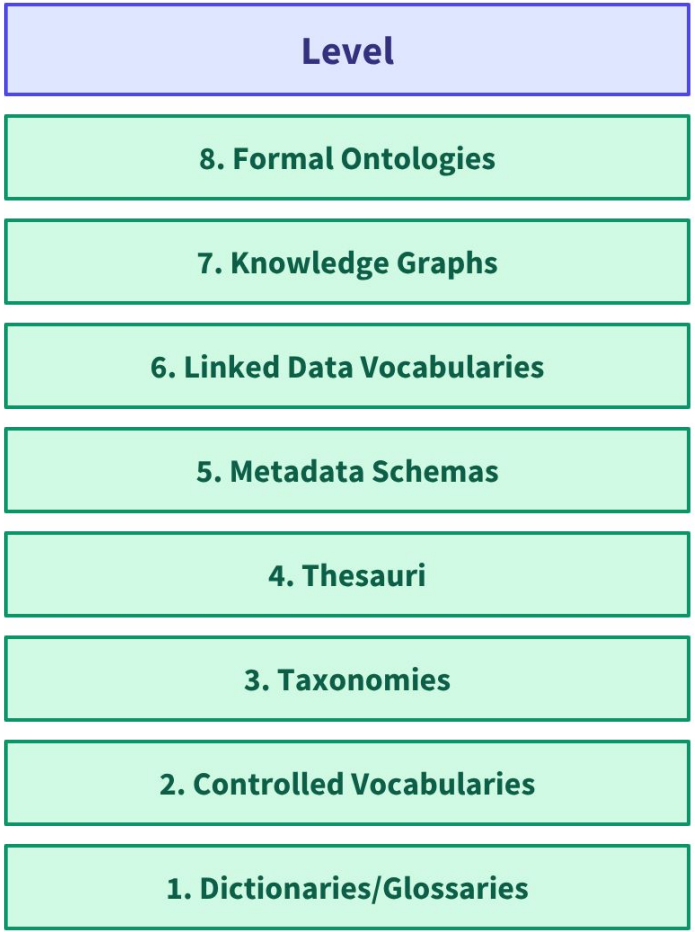
Be Meaningful

Vocabulary
Vocabularies can range from term list to ontology.

A term list tells agents what words you use. A taxonomy adds broader and narrower terms, and an ontology adds definitions and relationships.

Even an informal data dictionary helps.

There are a range of tools for getting started with development of this type of resource.



Be Meaningful

Structured Metadata Profiles

Leverage existing schema and profiles where you can.

All but DataCite leverage schema.org and all leverage a range of vocabularies to address things like prov, spatial etc.

Declare which profile you follow with `conformsTo`, so agents and validators know what to expect.

Others of potential interest

- Science on Schema.org

Profile	What it adds	Use it for
DataCite Metadata Schema 4.6 Custom schema with mappings	DOI-backed citation, creators, funders, related identifiers, rights	Generalist repos (Zenodo, Figshare, Dryad, Dataverse), following NIH GREI recommendations
Bioschemas Dataset	Life-science minimum, recommended, and optional fields; ontology-backed keywords	Life-science discovery and validation
CODAT CDIF	a set of profiles that address the most important functions for cross-domain FAIR implementation by providing core metadata fields	A framework to address a range of interoperability approaches in a FAIR context.
NIAID Data Ecosystem schema	IID fields: infectious agent, health condition, species, funding Based on BioSchemas	Harvest into the NDE Discovery Portal
Croissant 1.1 (+ RAI and Geo)	Files, typed record fields, ML splits, provenance, usage policies, responsible-AI fields	Broad SI workflow use with a focus on ML tools and workflows.
RO-Crate 1.2	Packages data, code, and workflow runs with their provenance	Data packaging with focus on reproducibility and workflows (Galaxy, WorkflowHub)

Be Meaningful

Let's revisit:

- **Rights & Access**
- Attribution and Authority
- Lineage and IP Protection

We can not guarantee we can address these, but we can do what we can and place the burden on others.

Rights expression via SPDX links
https://schema.org/usageInfo
https://schema.org/license
https://schema.org/acquireLicensePage

Access is evolving
Data Privacy Vocabulary (https://w3c-cg.github.io/dpv/2.0/)

Example SPDX links in schema.org

```
{  "@context": "https://schema.org/",
  "@id": "http://www.sample-data-repository.org/dataset/123",
  "@type": "Dataset",
  "name": "Removal of organic carbon by natural bacterioplankton
communities as a function of pCO2 from laboratory experiments between 2012
and 2016",
  "license": [ "http://spdx.org/licenses/CC0-1.0",
"https://creativecommons.org/publicdomain/zero/1.0"]
  ...
```

Be Meaningful

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Is defined by PIDs and Roles (for Authority)

Leverage something like NISO credit taxonomy (https://credit.niso.org/) and schema.org

```
{
  "@type": "Role",
  "roleName": "Conceptualization",
  "url":
"https://credit.niso.org/contributor-roles/conceptual
ization/",
  "contributor": { "@id":
"https://orcid.org/0000-..." }
}
```

Be Meaningful

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- Rights & Access
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- **Lineage and IP Protection**

We can not guarantee we can address these, but we can do what we can and place the burden on others.

W3C PROV-O (Provenance Ontology):
W3C standard model for entities, activities, and agents (`wasGeneratedBy`, `wasDerivedFrom`, `wasAssociatedWith`, etc.)

- **Versioning and relations**:
Use DataCite version properties and RelatedIdentifiers;
  - `IsVersionOf`
  - `IsNewVersionOf`
  - `IsDerivedFrom`
  - etc.

Link to upstream sources and downstream products.

- **Additional mechanisms**:
  - Leverage checksums or digital signatures on packages;

Be Meaningful

Caution: bleeding edge



A deeper set of context elements

Validation with SHACL Core (a W3C Recommendation). A SHACL shape is a machine-readable rule for what a valid record looks like, such as "every sample has a collection date and a host taxon." (https://w3c.github.io/data-shapes/shacl/)

Rules with SPARQL RL (a W3C Working Draft, Oct 21 2026). Rules derive new facts, such as computing a value or inferring a parent category, so curation knowledge is declared rather than hidden in code. (https://www.w3.org/TR/sparql12-rl/)

Mappings with SSSOM. SSSOM is a simple table format (CSV) for saying "my term X matches term Y over there." Each row records the relation (exact, close, broader or related), the justification and who asserted it, so an agent can cross between vocabularies with known confidence. (https://mapping-commons.github.io/sssom/dev/)

Exposing these to agents, especially when coupled with a Knowledge Graph, provides very expressive use of that KG

Sharing and aligning your context elements:

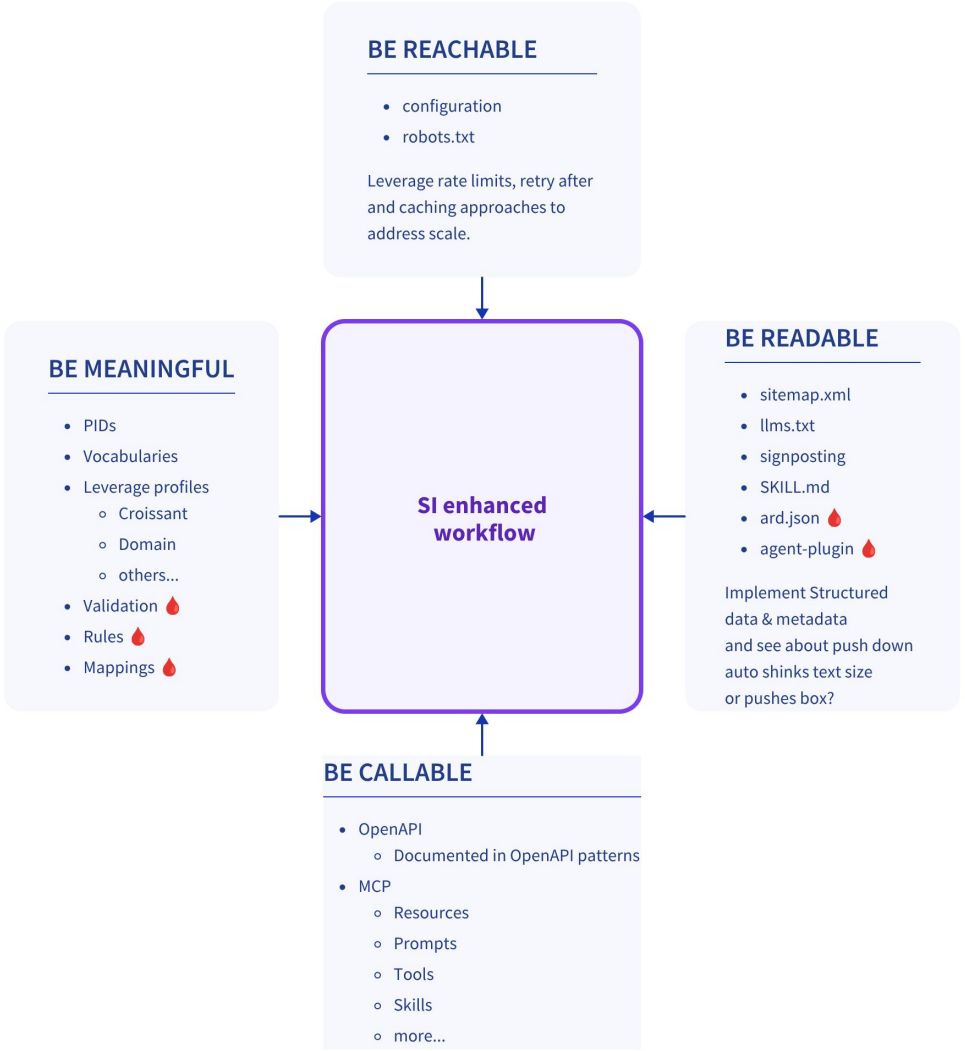
- Facilitates *harmonization and integration* across facilities
- Enables entity resolution and relation extraction pipelines

Be All That You Can Be

Facilitating the SI workflow of your users.

- ***Reachable*** *(network accessible)*
- ***Readable*** *(expressive resources)*
- ***Callable*** *(dynamic/service reads)*
- ***Meaningful*** *(context)*

Provide knowledge based determinism to the generative engine.



Evaluation

Evaluation is a huge topic, but where can we start?

First, some core questions for the community:

- How to **establish baselines**
- How to **measure success over time**

Caution: Too Big for Today 🏔️

Some tools you can you start with now:

- NDE Portal: Compare results with AI-assisted search on and off (https://data.niaid.nih.gov/knowledge-center/ai-assisted-search)
- Chrome Lighthouse's agent audit
  a. https://developer.chrome.com/docs/lighthouse
  b. Llms.txt Chrome lighthouse audits this https://developer.chrome.com/docs/lighthouse/agentic-browsing/llms-txt
  c. ard.json analyser (see also https://github.com/ards-project/ard-spec)

Working on some ideas for the liaison meeting on self assessment with custom skills based off this presentation.

A Closing Observation

Other than the APIs and MCP services, ***almost everything presented here is a static text file*** exposed by your server.

That doesn't mean it's easy,
but it does make it accessible.



Example Roadmap:

1. **Phase 1 (Basic):** `robots.txt`, `sitemap.xml`, `llms.txt`, basic Schema.org based profiles, PIDs
2. **Phase 2 (Intermediate):** OpenAPI described APIs, SKILL.md, Croissant/RO-Crate profiles (or other)
3. **Phase 3 (Advanced):** MCP services, OAuth 2.1 auth servers, SHACL/SSSOM semantic validation.

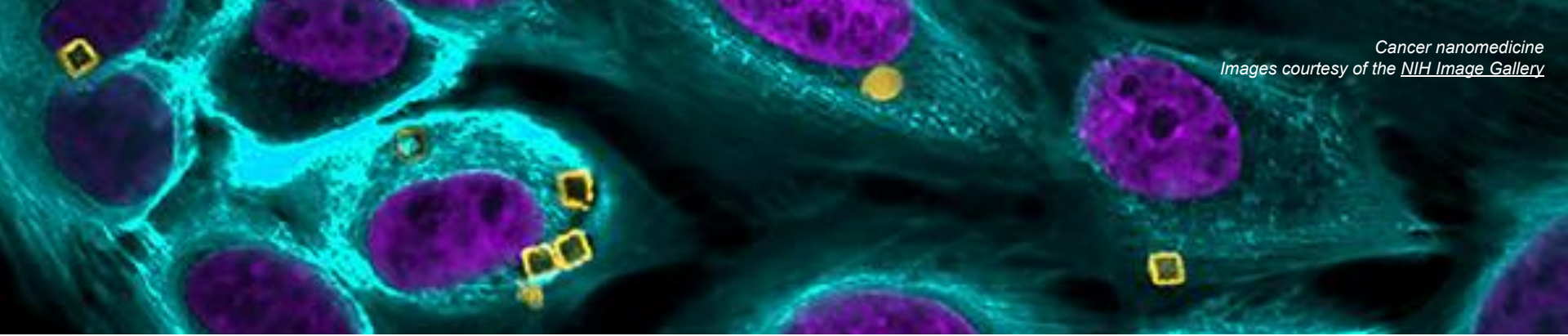


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Image source: https://www.niaid.nih.gov/



Cancer nanomedicine
Images courtesy of the NIH Image Gallery

Discussion

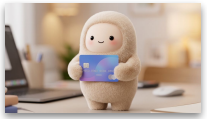
More From the Edge

Caution: bleeding edge



Things for your radar:

- Your Banks are dealing with the liability, human in the loop and many other factors around agents working on your behalf right now! It's worth watching that.
  *Go ahead… let Meta Muse have your credit card! **



- Exposing embeddings aligned to an agreed on model
- Connecting with Genesis Mission

- We describe our APIs, so why not our UIs?
  - Generative UIs
    - Open Intelligent UI
      - https://github.com/CopilotKit/OpenIntelligentUI
    - Opencode genui
      - https://github.com/madalapranavsai/genui-package
    - Herdr UI
      - https://herdr.dev/
    - Many more coming down the pipe…
  - WebMCP (Still trying to figure out the compelling use case for this one!)
    - https://webmachinelearning.github.io/webmcp/



* *Doug is not a financial advisor and this is not financial advice*